



Gender Differences in Time Spent on Education

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Introduction/Background

- Studies and news articles have highlighted how boys are falling behind girls in school with
 - lower reading scores (New York Times)
 - a graduation rate of 83% for boys and 89% for girls in 2023 (Brookings)
 - 66% women enrolled in college compared to 57% of men in 2022 (National Center for Education Statistics).

To investigate if this trend manifests itself in how much time students spend on education, I seek to answer the following questions:

Has time spent on educational activities varied by gender from 2003-2024? Are there other factors, including income, location, race or their interactions with gender that contribute to differences in time spent on education?

Data

Data Sources

- Roster, Respondent, Activity, and Current Population Survey files from the American Time Use Survey Data (ATUS) spanning 2003-2024.

Sample

- 16,898 full-time students aged 15-24 in the ATUS dataset

Data Cleaning

- To define educational activities, I included all categories under the major category of education with the label 06. The main activities include taking class, extracurricular club activities excluding sports, and homework.
- I summed all the time people spent on educational activities in a day, and filled in 0's for people who had no recorded time spent on education.
- I then aggregated several categorical variables into broader categories to avoid having too many levels, including income to 5 levels, race to 6 levels, and year into 4 periods of 4 years each with the last period being 6 years.

Models

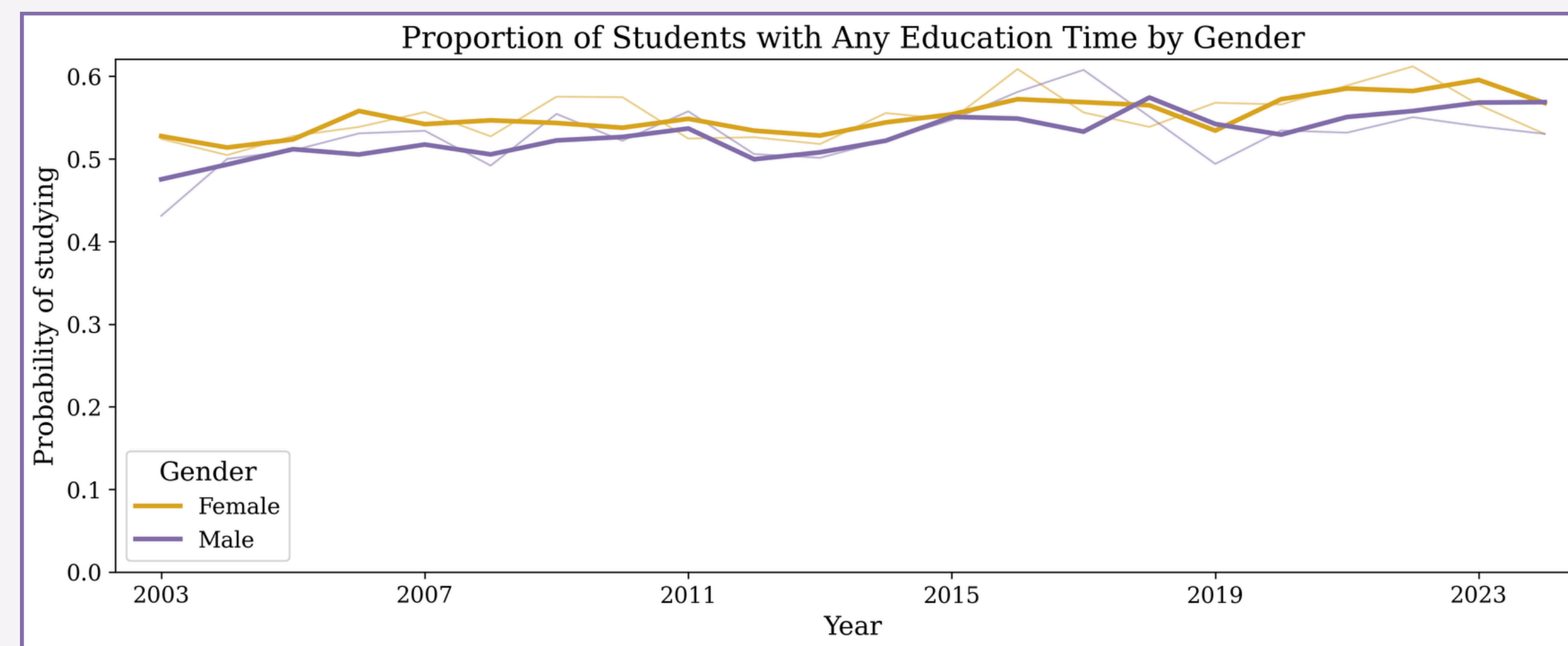
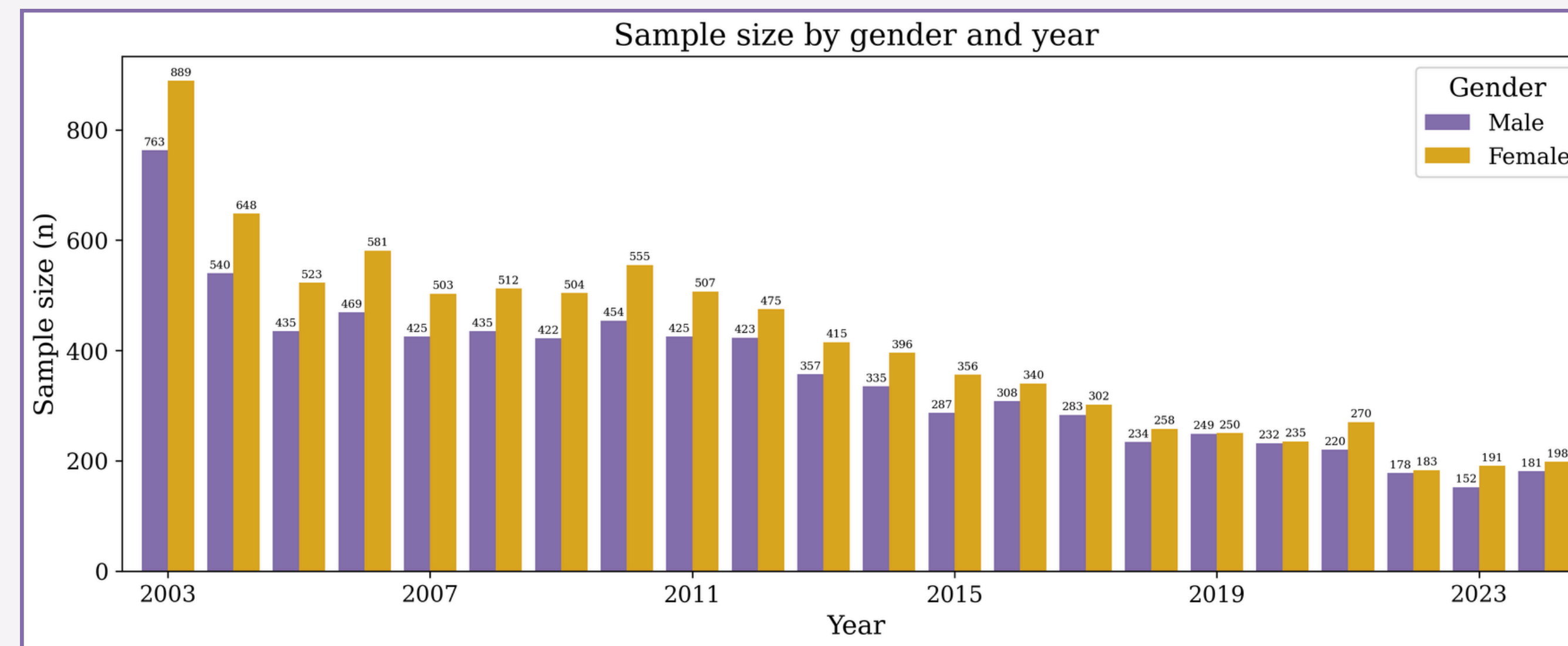
Predictors

- Primary Predictor: Gender
- Secondary Predictors: Household Family Income, Race, Region
- Interactions w/ gender: Race, Income, Region, Year, College vs. High School
- Other predictors to control for: Weekday vs. Weekend, Month, Period, College vs. High School

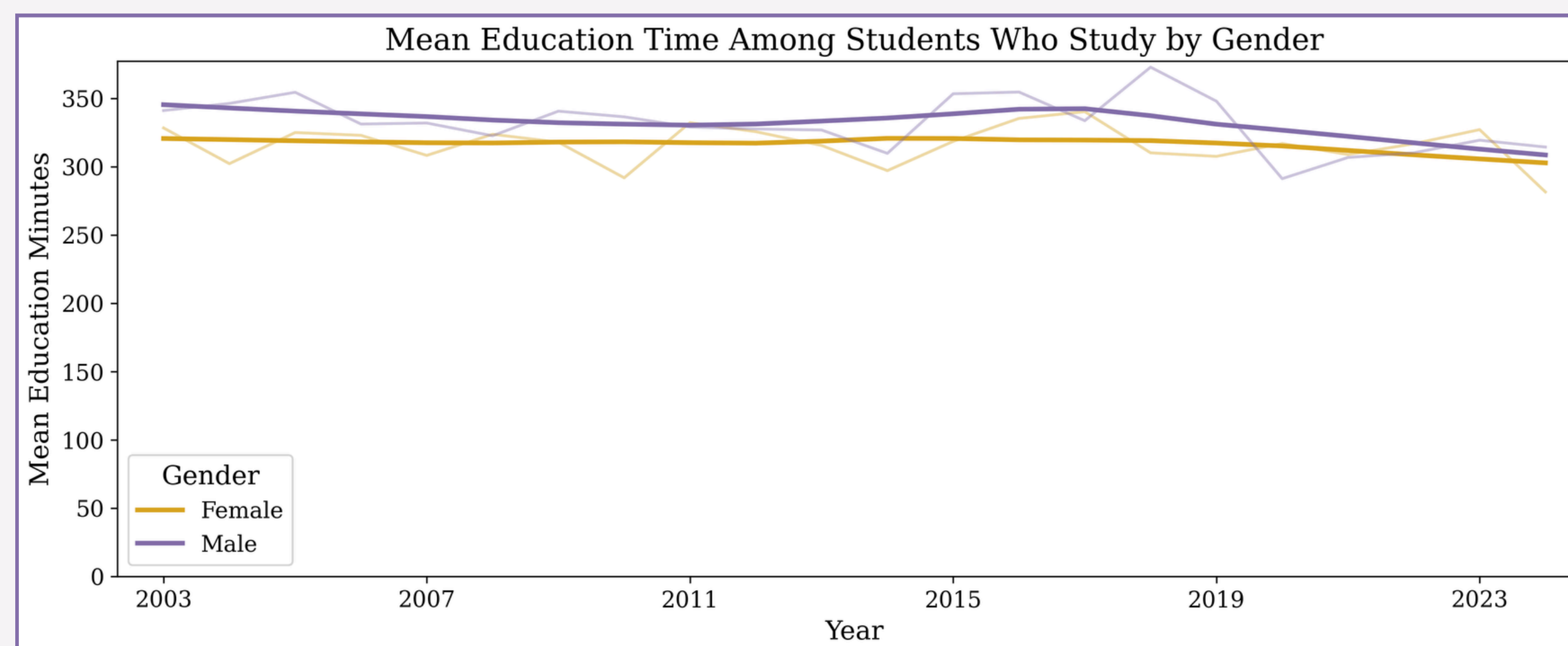
- 4 models: 2 logistic regression models and 2 linear regression models**
- Each model type is run twice with:
 - A focus on gender differences with the variable period (aggregated years).
 - An interaction between numeric year and gender to see if there is change over time by gender.

- Logistic Regression Model Outcome:** Whether or not the student spent any time on educational activities at all
- Linear Regression Model Outcome:** For students who spent any time on education (n = 9024), the logged minutes spent on educational activities

Visualizations



Note: Lighter lines = actual proportions and darker lines = predicted probabilities



Note: Lighter lines = actual means and darker lines = predicted means

Results

Significant Predictors:

- Logistic Regression**
- Model 1: Gender, Indicator for Asian and Other category for race, Indicators for middle low and high income, Indicator for College, Weekend, months besides May, periods, interaction between female and in college
- Model 2: Year, middle income & the same predictors as above w/o gender
- Linear Regression**
- Model 1 and 2:** Indicator for Asian, Indicator for middle and middle high, Indicator for College, Weekend, June-August, October, the interaction of female and hispanic, female and middle high income, & female and college

Logistic Regression	Odds Ratio	P-Value	95% Confidence Interval
Female (first model)	1.435	0.003	(1.127, 1.828)
Asian	1.722	< 0.0001	(1.394, 2.133)
Female x College	0.845	0.016	(0.74, 0.97)
2008-2012	1.12	0.009	(1.028, 1.219)
2013-2017	1.215	< 0.0001	(1.105, 1.337)
2018-2024	1.165	0.003	(1.054, 1.288)

Linear Regression	Coefficient Estimate	P-Value	95% Confidence Interval
Female	-8.04	0.145	(-17.66, 2.71)
Asian	20.78	< 0.0001	(10.69, 31.79)
Weekend	-53.98	< 0.0001	(-55.65, -52.24)
Female x Hispanic	9.83	0.04	(0.29, 20.29)

Multiple R-squared: 0.21

Discussion and Conclusion

- There is **a statistically significant difference between boys and girls in whether or not they spend time on education at all**. Girls have a 43.5% higher odds of studying on their survey day compared to boys.
- However, there is no statistically significant gender gap in how much time students spend on educational activities if they did engage in them.
- Other predictors statistically significant in time spent on education include Asian students reporting a higher likelihood and minutes studying.

Limitations

- ATUS sample size is decreasing across years, which possibly affects the year variation change in education time.
- Near equal amount of weekends and weekdays in the ATUS dataset-weekend overrepresentation and less time spent on educational activities.
- Family Income is grouped into fixed buckets
- Mild heteroskedasticity for the linear regression model, as there is a lot of variability in educational time across levels of independent variables
 - Handled by HC3, a type of heteroskedasticity consistent robust standard errors, which manage high leverage observations and is used for small- medium sample sizes

Future work

- Disaggregate Asian by specific race
- Explore other interactions, such as interactions for race and income

References

- U.S. Bureau of Labor Statistics. (2024). American Time Use Survey (ATUS). <https://www.bls.gov/tus/>
- Miller, C. C. (2025, May 13). It's not just a feeling: Data shows boys and young men are falling behind. The New York Times. <https://www.nytimes.com/2025/05/13/upshot/boys-falling-behind-data.html>
- Reeves, R. V., & Smith, E. (2023, April 18). Racial disparities in the high school graduation gender gap. Brookings Institution. <https://www.brookings.edu/articles/racial-disparities-in-the-high-school-graduation-gender-gap/>
- National Center for Education Statistics. (2023). Digest of education statistics 2023, Table 302.10: Number of recent high school completers and percent enrolled in college, by sex and level of institution (1960-2022). U.S. Department of Education. https://nces.ed.gov/programs/digest/d23/tables/dt23_302.10.asp